

Integral Calculus (2015–2021)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

32

TOTAL MARKS

54 M

TEST DURATION

96 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	16	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	2	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	14	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q43 • ID: JAM_2021_Q43)

NAT

+1 M

Neg: Nil (0)

Q. 43 Let $B = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$ and define $u(x, y, z) = \sin((1 - x^2 - y^2 - z^2)^2)$ for $(x, y, z) \in B$. Then the value of

$$\iiint_B \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) dx dy dz$$

is _____.

Question 2 (JAM 2021 • Q5 • ID: JAM_2021_Q5)

MCQ

+1 M

Neg: -0.33

Q. 5 Let p and t be positive real numbers. Let D_t be the closed disc of radius t centered at $(0, 0)$, i.e., $D_t = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq t^2\}$. Define

$$I(p, t) = \iint_{D_t} \frac{dx dy}{(p^2 + x^2 + y^2)^p}.$$

Then $\lim_{t \rightarrow \infty} I(p, t)$ is finite

(A) only if $p > 1$.

(B) only if $p = 1$.

(C) only if $p < 1$.

(D) for no value of p .

Question 3 (JAM 2020 • Q47 • ID: JAM_2020_Q47)

NAT

+1 M

Neg: Nil (0)

Q. 47 If $\int_0^1 \int_{2y}^2 e^{x^2} dx dy = k(e^4 - 1)$, then k equals _____.

Question 4 (JAM 2020 • Q38 • ID: JAM_2020_Q38)

MSQ

+2 M

Neg: Nil (0)

Q. 38 Let S be that part of the surface of the paraboloid $z = 16 - x^2 - y^2$ which is above the plane $z = 0$ and D be its projection on the xy -plane. Then the area of S equals

(A) $\iint_D \sqrt{1 + 4(x^2 + y^2)} dx dy$

(B) $\iint_D \sqrt{1 + 2(x^2 + y^2)} dx dy$

(C) $\int_0^{2\pi} \int_0^4 \sqrt{1 + 4r^2} r dr d\theta$

(D) $\int_0^{2\pi} \int_0^4 \sqrt{1 + 4r^2} r dr d\theta$

Question 5 (JAM 2020 • Q25 • ID: JAM_2020_Q25)

MCQ

+2 M

Neg: -0.67

Q. 25 Let S be the part of the cone $z^2 = x^2 + y^2$ between the planes $z = 0$ and $z = 1$. Then the value of the surface integral $\iint_S (x^2 + y^2) dS$ is

(A) π

(B) $\frac{\pi}{\sqrt{2}}$

(C) $\frac{\pi}{\sqrt{3}}$

(D) $\frac{\pi}{2}$

MA

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Question 6 (JAM 2020 • Q24 • ID: JAM_2020_Q24)

MCQ

+2 M

Neg: -0.67

Q. 24 The value of the triple integral $\iiint_V (x^2 y + 1) dx dy dz$, where V is the region given by $x^2 + y^2 \leq 1, 0 \leq z \leq 2$ is

(A) π

(B) 2π

(C) 3π

(D) 4π

Question 7 (JAM 2020 • Q18 • ID: JAM_2020_Q18)

MCQ

+2 M

Neg: -0.67

Q. 18 The area bounded by the curves $x^2 + y^2 = 2x$ and $x^2 + y^2 = 4x$, and the straight lines $y = x$ and $y = 0$ is

(A) $3 \left(\frac{\pi}{2} + \frac{1}{4} \right)$

(B) $3 \left(\frac{\pi}{4} + \frac{1}{2} \right)$

(C) $2 \left(\frac{\pi}{4} + \frac{1}{3} \right)$

(D) $2 \left(\frac{\pi}{3} + \frac{1}{4} \right)$

Question 8 (JAM 2020 • Q16 • ID: JAM_2020_Q16)

MCQ

+2 M

Neg: -0.67

JAM 2020

MATHEMATICS - MA

Q. 16 Let S be the surface of the portion of the sphere with centre at the origin and radius 4, above the xy -plane. Let $\vec{F} = y\hat{i} - x\hat{j} + yx^3\hat{k}$. If $\hat{n}$ is the unit outward normal to S , then

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS$$

equals

(A) -32π (B) -16π (C) 16π (D) 32π

Question 9 (JAM 2019 • Q51 • ID: JAM_2019_Q51)

NAT

+2 M

Neg: Nil (0)

Q. 51 – Q. 60 carry two marks each.

Q.51 The volume of the solid of revolution of the loop of the curve $y^2 = x^4(x + 2)$ about the x -axis (round off to 2 decimal places) is _____

MA

11/13

Question 10 (JAM 2019 • Q50 • ID: JAM_2019_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50 The volume of the solid bounded by the surfaces $x = 1 - y^2$ and $x = y^2 - 1$, and the planes $z = 0$ and $z = 2$ (round off to 2 decimal places) is _____

Q. 51 – Q. 60 carry two marks each

Question 11 (JAM 2019 • Q49 • ID: JAM_2019_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49 The value of the integral

$$\int_{-1}^1 \int_{-1}^1 |x + y| \, dx \, dy$$

(round off to 2 decimal places) is _____

Question 12 (JAM 2019 • Q23 • ID: JAM_2019_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 The area of the part of the surface of the paraboloid $x^2 + y^2 + z = 8$ lying inside the cylinder $x^2 + y^2 = 4$ is

- (A) $\frac{\pi}{2}(17^{3/2} - 1)$ (B) $\pi(17^{3/2} - 1)$ (C) $\frac{\pi}{6}(17^{3/2} - 1)$ (D) $\frac{\pi}{3}(17^{3/2} - 1)$

Question 13 (JAM 2019 • Q10 • ID: JAM_2019_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 The area of the surface generated by rotating the curve $x = y^3$, $0 \leq y \leq 1$, about the y-axis, is

- (A) $\frac{\pi}{27} 10^{3/2}$ (B) $\frac{4\pi}{3} (10^{3/2} - 1)$ (C) $\frac{\pi}{27} (10^{3/2} - 1)$ (D) $\frac{4\pi}{3} 10^{3/2}$

Q. 11 – Q. 30 carry two marks each.

Question 14 (JAM 2019 • Q9 • ID: JAM_2019_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 The value of the integral

$$\int_{y=0}^1 \int_{x=0}^{1-y^2} y \sin(\pi(1-x)^2) \, dx \, dy$$

is

- (A) $\frac{1}{2\pi}$ (B) 2π (C) $\frac{\pi}{2}$ (D) $\frac{2}{\pi}$

Question 15 (JAM 2018 • Q58 • ID: JAM_2018_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58 The area of the parametrized surface

$$S = \{((2 + \cos u) \cos v, (2 + \cos u) \sin v, \sin u) \in \mathbb{R}^3 \mid 0 \leq u \leq \frac{\pi}{2}, 0 \leq v \leq \frac{\pi}{2}\}$$

is _____ (correct up to two decimal places).

Question 16 (JAM 2018 • Q56 • ID: JAM_2018_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 The value of the integral

$$\int_0^1 \int_x^1 y^4 e^{xy^2} dy dx$$

is _____ (correct up to three decimal places).

Question 17 (JAM 2018 • Q54 • ID: JAM_2018_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54 If the volume of the solid in $\mathbb{R}^3$ bounded by the surfaces

$$x = -1, \quad x = 1, \quad y = -1, \quad y = 1, \quad z = 2, \quad y^2 + z^2 = 2$$

is $\alpha - \pi$, then $\alpha =$ _____

Question 18 (JAM 2018 • Q19 • ID: JAM_2018_Q19)

MCQ

+2 M

Neg: -0.67

Q.19 Consider the region D in the yz plane bounded by the line $y = \frac{1}{2}$ and the curve $y^2 + z^2 = 1$, where $y \geq 0$. If the region D is revolved about the z -axis in $\mathbb{R}^3$, then the volume of the resulting solid is

(A) $\frac{\pi}{\sqrt{3}}$

(B) $\frac{2\pi}{\sqrt{3}}$

(C) $\frac{\pi\sqrt{3}}{2}$

(D) $\pi\sqrt{3}$

Question 19 (JAM 2017 • Q56 • ID: JAM_2017_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 For a real number x , define $[x]$ to be the smallest integer greater than or equal to x . Then

$$\int_0^1 \int_0^1 \int_0^1 ([x] + [y] + [z]) \, dx \, dy \, dz = \underline{\hspace{2cm}}$$

Question 20 (JAM 2017 • Q32 • ID: JAM_2017_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32 The volume of the solid

$$\left\{ (x, y, z) \in \mathbb{R}^3 \mid 1 \leq x \leq 2, \quad 0 \leq y \leq \frac{2}{x}, \quad 0 \leq z \leq x \right\}$$

is expressible as

(A) $\int_1^2 \int_0^{2/x} \int_0^x dz \, dy \, dx$

(B) $\int_1^2 \int_0^x \int_0^{2/x} dy \, dz \, dx$

(C) $\int_0^2 \int_1^z \int_0^{2/x} dy \, dx \, dz$

(D) $\int_0^2 \int_{\max\{z, 1\}}^2 \int_0^{2/x} dy \, dx \, dz$

Question 21 (JAM 2017 • Q16 • ID: JAM_2017_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 The area of the surface $z = \frac{xy}{3}$ intercepted by the cylinder $x^2 + y^2 \leq 16$ lies in the interval

(A) $(20\pi, 22\pi]$

(B) $(22\pi, 24\pi]$

(C) $(24\pi, 26\pi]$

(D) $(26\pi, 28\pi]$

Question 22 (JAM 2017 • Q6 • ID: JAM_2017_Q6)

MCQ

+1 M

Neg: -0.33

Q.6

$$\int_0^1 \int_x^1 \sin(y^2) \, dy \, dx =$$

(A) $\frac{1+\cos 1}{2}$

(B) $1 - \cos 1$

(C) $1 + \cos 1$

(D) $\frac{1-\cos 1}{2}$

Question 23 (JAM 2016 • Q57 • ID: JAM_2016_Q57)**NAT****+2 M****Neg: Nil (0)**

Q.57 The value of the double integral

$$\int_0^{\pi} \int_0^x \frac{\sin y}{\pi - y} dy dx$$

is _____

Question 24 (JAM 2016 • Q52 • ID: JAM_2016_Q52)**NAT****+2 M****Neg: Nil (0)**Q.52 Let R be the region enclosed by $x^2 + 4y^2 \geq 1$ and $x^2 + y^2 \leq 1$. Then the value of

$$\iint_R |xy| dx dy \text{ is } \underline{\hspace{2cm}}$$

Question 25 (JAM 2016 • Q16 • ID: JAM_2016_Q16)**MCQ****+2 M****Neg: -0.67**Q.16 The surface area of the portion of the plane $y + 2z = 2$ within the cylinder $x^2 + y^2 = 3$ is

(A) $\frac{3\sqrt{5}}{2}\pi$

(B) $\frac{5\sqrt{5}}{2}\pi$

(C) $\frac{7\sqrt{5}}{2}\pi$

(D) $\frac{9\sqrt{5}}{2}\pi$

Question 26 (JAM 2016 • Q15 • ID: JAM_2016_Q15)

MCQ

+2 M

Neg: -0.67

Q.15 If the triple integral over the region bounded by the planes

$$2x + y + z = 4, \quad x = 0, \quad y = 0, \quad z = 0$$

is given by

$$\int_0^2 \int_0^{\lambda(x)} \int_0^{\mu(x,y)} dz \, dy \, dx,$$

then the function $\lambda(x) - \mu(x, y)$ is

- (A) $x + y$ (B) $x - y$ (C) x (D) y

Question 27 (JAM 2015 • Q58 • ID: JAM_2015_Q58)

NAT

+2 M

Neg: Nil (0)

Q.18

Let ℓ be the length of the portion of the curve $x = x(y)$ between the lines $y = 1$ and $y = 3$, where $x(y)$ satisfies

$$\frac{dx}{dy} = \frac{\sqrt{1 + y^2 + y^4}}{y}, \quad x(1) = 0$$

The value of ℓ , correct upto three decimal places, is _____

Question 28 (JAM 2015 • Q55 • ID: JAM_2015_Q55)

NAT

+2 M

Neg: Nil (0)

Q.15

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{y}{\sin y}, & y \neq 0 \\ 1, & y = 0 \end{cases}$$

Then the integral

$$\frac{1}{\pi^2} \int_{x=0}^1 \int_{y=\sin^{-1} x}^{\frac{\pi}{2}} f(x, y) \, dy \, dx$$

correct upto three decimal places, is _____

Question 29 (JAM 2015 • Q42 • ID: JAM_2015_Q42)

NAT

+1 M

Neg: Nil (0)

Q.2

Let S be the portion of the surface $z = \sqrt{16 - x^2}$ bounded by the planes $x = 0$, $x = 2$, $y = 0$, and $y = 3$. The surface area of S , correct upto three decimal places, is _____

Question 30 (JAM 2015 • Q26 • ID: JAM_2015_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 The area of the planar region bounded by the curves $x = 6y^2 - 2$ and $x = 2y^2$ is

(A) $\frac{\sqrt{2}}{3}$

(B) $\frac{2\sqrt{2}}{3}$

(C) $\frac{4\sqrt{2}}{3}$

(D) $\sqrt{2}$

Question 31 (JAM 2015 • Q14 • ID: JAM_2015_Q14)

MCQ

+2 M

Neg: -0.67

Q.14 Let S be the bounded surface of the cylinder $x^2 + y^2 = 1$ cut by the planes $z = 0$ and $z = 1 + x$. Then the value of the surface integral $\iint_S 3z^2 d\sigma$ is equal to

(A) $\int_0^{2\pi} (1 + \cos \theta)^3 d\theta$

(B) $\int_0^{2\pi} \sin \theta \cos \theta (1 + \cos \theta)^2 d\theta$

(C) $\int_0^{2\pi} (1 + 2 \cos \theta)^3 d\theta$

(D) $\int_0^{2\pi} \sin \theta \cos \theta (1 + 2 \cos \theta)^2 d\theta$

Question 32 (JAM 2015 • Q6 • ID: JAM_2015_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 The volume of the portion of the solid cylinder $x^2 + y^2 \leq 2$ bounded above by the surface $z = x^2 + y^2$ and bounded below by the xy -plane is

(A) π

(B) 2π

(C) 3π

(D) 4π

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Integral Calculus (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+1	0
2	MCQ	+1	A
3	NAT	+1	0.25 to 0.25
4	MSQ	+2	A;D
5	MCQ	+2	B
6	MCQ	+2	B
7	MCQ	+2	B
8	MCQ	+2	A
9	NAT	+2	6.60 to 6.80
10	NAT	+1	5.30 to 5.50
11	NAT	+1	2.60 to 2.70

Q. No.	Type	Marks	Official Key
12	MCQ	+2	C
13	MCQ	+1	C
14	MCQ	+1	A
15	NAT	+2	6.30 to 6.70
16	NAT	+2	0.230 to 0.250
17	NAT	+2	5.99 to 6.01
18	MCQ	+2	C
19	NAT	+2	2.9 to 3.1
20	MSQ	+2	A, B, D
21	MCQ	+2	A
22	MCQ	+1	D

Q. No.	Type	Marks	Official Key
23	NAT	+2	2.0 to 2.0
24	NAT	+2	0.35 to 0.4
25	MCQ	+2	A
26	MCQ	+2	D
27	NAT	+2	5.09 to 5.10
28	NAT	+2	0.125
29	NAT	+1	6.28 to 6.29
30	MCQ	+2	C
31	MCQ	+2	A
32	MCQ	+1	B

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.